RÉSUMÉ

Ivan Ball-Ilovera

Principal Software Architect

Professional summary

Principal Software Architect with 25+ years designing and delivering scalable, cloud-native systems on the Microsoft stack. Deep expertise in Domain-Driven Design, Clean Architecture, CQRS, and microservices, paired with hands-on Azure cloud architecture and DevOps (Bicep, Docker, Kubernetes, CI/CD). Proven record of leading architectural refactors, modernizing legacy platforms, and mentoring cross-functional teams to raise engineering standards. Currently architecting production-grade .NET 10 platforms spanning modular-monolith-to-microservices, event-driven messaging, and AI/RAG integration, while leveraging AI-assisted development workflows to ship faster without sacrificing quality.

Core competencies

Architecture & design	Domain-Driven Design, Clean Architecture, CQRS, Modular Monolith → Microservices, Event-Driven Architecture, Outbox Pattern, gRPC, API Gateway (YARP), design patterns.
Cloud & DevOps	Azure (Container Apps, Service Bus, Cosmos DB, SQL, App Insights, Log Analytics), Infrastructure as Code (Bicep, ARM), Docker, Kubernetes, Service Fabric, CI/CD (GitHub Actions, Azure DevOps), OIDC / Managed Identity.
Languages & frameworks	.NET (Core through 10) / C#, Blazor, .NET MAUI, ASP.NET Core Web API, .NET Aspire, EF Core, Angular, TypeScript, Python.
Data & BI	SQL Server, Cosmos DB, MySQL, Oracle, data engineering, data warehousing, SSIS, SSAS (Tabular), SSRS, Power BI.
Quality & security	xUnit, NUnit, Moq, Playwright, Selenium, TDD, static analysis and application security (SAST/DAST, Snyk, Fortify, Wiz).
Leadership & practices	Technical leadership, mentorship, Agile / Scrum / Kanban, architecture governance (ADRs), cross-team collaboration, AI-assisted development (Claude Code, GitHub Copilot).

Professional experience

Senior Software Engineer · Assurant

`June 2025 – Present · Architect-level scope: platform, security, and cross-team technical decisions`

- Re-architected the AR.com renters quote and issue flow for multi-product quoting (Renters Insurance plus General Liability) against Guidewire, delivering Orchestration and Sales API changes, Angular UI updates, reusable producer-code propagation, and a new guidewireData integration contract that enabled the modern Guidewire system-of-record path end to end.
- Delivered Okta-based multi-factor authentication (SMS and voice, magic-link, and sign-in widget) across all FlexDeposit flows (payment, search-bond, tenant-portal, and extension), with OTP retry, resend throttling, and account lockout.
- Led the Angular 18 to 21 major-version upgrade and hardened application security: remediated 9 high-severity and 12 medium and low Wiz vulnerabilities, ran Snyk scans across the UI and Places API, and completed a full Fortify-to-Snyk SAST migration.
- Led the migration of the AR.com repositories from Azure DevOps to GitHub Enterprise with full history preserved, re-pointing CI/CD pipelines, branch protection, and team permissions, which unlocked GitHub Copilot PR review.
- Designed and shipped branch-level coverage sync (Azure Data Factory, API, and SQL) that resolved a production data-integrity incident in which Model-to-Prod syncs were silently overwriting production coverage changes.
- Architected AI delivery tooling in two company hackathons: a Clean Architecture and machine-learning solution for automated insurance-claim fraud detection, and "A.I.M." (AI Initiative & Modernization), an AI assistant embedded in Azure DevOps that spans the full delivery lifecycle (plan, build, test, ship, operate) with a multi-provider, human-in-the-loop design; adopted GitHub Copilot and Claude Code to raise code quality and developer efficiency.

Principal Software Engineer · Hitachi Solutions America

May 2021 – May 2025

- Led the architectural refactor of biBERK's cloud-native insurance platform (C#, Angular, Azure SQL, Cosmos DB), modernizing the system and establishing scalable patterns adopted across the organization.
- Served as the go-to architect across biBERK, Empower, Analog Devices, and Cost Management engagements; coached engineers on methods, tooling, and best practices.
- Established the Empower front-end foundation on Angular best practices and authored cross-cutting design specifications spanning UI, API, and database.
- Built CI/CD pipelines using Infrastructure as Code (ARM); developed a Python document-ingestion module for a Retrieval-Augmented Generation (RAG) architecture powering internal AI initiatives.

Software Engineer IV · Ultimate Software

December 2015 – May 2021

- Architected a PostUpgrade automated testing engine (.NET Core, Angular, Selenium) and defined the architecture for the CAT application suite, heading off ~32 customer escalations and saving an estimated 80 hours of support time.
- Led the migration of 11 production database shards from MariaDB (Pivotal Cloud Foundry) to MySQL on Kubernetes, adding Slack-integrated monitoring and alerting.
- Re-engineered WFM ETL processing (SSIS) across all data centers, cutting execution time from 2+ hours to under an hour.
- Built the SpecParser app and a nightly failure report that surfaced upgrade/config one-off bugs early; established engineering best practices within Agile/Scrum.

Senior Software Architect, Lead Developer & DBA · Miami-Dade County (Property Appraiser's Office)

August 2009 – December 2015

- Built a reusable .NET application framework that enabled a team to deliver line-of-business applications (ASP.NET Web Forms N-Tier, then ASP.NET MVC, HTML5/CSS3/JavaScript/Bootstrap, Entity Framework 6).
- Architected the SQL Server backend for the County's most-trafficked web app (Property Search, 250,000+ hits/day), using advanced partitioning across multi-cluster, load-balanced servers.
- Developed a flexible security module with dynamic authentication/authorization, integrated with HR systems and Active Directory via LDAP.
- Implemented an enterprise data warehouse loading SQL Server from multiple Oracle sources (re-architected from a multithreaded C# loader to SSIS), surfacing analytics through an SSAS Tabular/multidimensional model and SSRS reports.

Earlier experience

1999 – 2009

- Progressive software engineering and architecture roles delivering line-of-business, ERP, and real-estate-appraisal systems across the Microsoft and Borland stacks: Advantage Integration Systems, Software Techniques, Mariner Systems, Arthur Douglas & Associates, and Thales Information Systems (ERP "Alquimia" in Borland JBuilder X + Oracle 9i).
- Began a software engineering career at DataCIMEX in Havana, Cuba (1999 – 2004), developing business applications and database-backed systems.

Featured project · MMCA platform

Personal / open source · github.com/ivanball/MMCA.Common

A production-grade .NET 10 reference platform demonstrating modern enterprise architecture end-to-end. The conference application was deployed to Azure and powered the live 2026 Atlanta Cloud + AI Conference, and will also power the 2026 Atlanta Developers Conference.

- Designed a modular monolith that extracts cleanly into independent microservices behind a YARP gateway, with gRPC contracts and typed clients keeping transport reversible at the edge.
- Authored a 22-package shared framework (DDD, Clean Architecture, CQRS) published to nuget.org and GitHub Packages and consumed by an e-commerce app (Stripe), a conference app, and a minimal reference app.
- Implemented database-per-service with an at-least-once outbox plus dual dispatch, soft-delete, automatic audit stamping, and GDPR/CCPA anonymization.
- Deployed to Azure with Bicep IaC, Service Bus, Container Apps, App Insights, and Log Analytics, using OIDC / managed-identity authentication; orchestrated locally with .NET Aspire.
- Built Blazor (MudBlazor) and .NET MAUI clients; enforced quality with xUnit v3, Playwright E2E, 141 architecture-fitness tests, and automated accessibility testing (axe, WCAG 2.1 AA: zero violations).
- Documented decisions with 132 ADRs and a 34-category architecture-review rubric; built Roslyn-based code-inventory tooling and AI/multi-agent development workflows.

Read the full platform breakdown →

Education

B.S., Computer Science
University of Havana (Faculty of Mathematics), Havana, Cuba
(1994 – 1999)

Languages

English · Spanish (bilingual)

Certifications

- ✓ **Azure Administrator Associate** (AZ-104, 2025)
- ✓ **Azure AI Fundamentals** (AI-900, 2024)
- ✓ **Azure Data Fundamentals** (DP-900, 2021)
- ✓ **Azure Fundamentals** (AZ-900, 2021)
- → In progress (2026): Azure Solutions Architect Expert (AZ-305), Azure AI Engineer Associate (AI-102), with AZ-400 and the GitHub Copilot certification to follow.

Earlier Microsoft certifications (retired programs, 2006 – 2016): MCSA SQL Server 2012, MCSA Web Applications (ASP.NET MVC), plus MCTS/MCPD .NET and Solution Architecture credentials.

Professional development

Continuously prototypes emerging technologies, with a current focus on Clean Architecture using .NET 10, Blazor, .NET MAUI, and ASP.NET Core Web API, and on AI-assisted development with Claude Code and GitHub Copilot to accelerate delivery and code quality. Stays engaged with the developer community through Pluralsight courses, user groups, and code camps.